Grid spacing decides which microphysical pathway controls the liquid water path response to droplet number in simulated drizzling stratocumulus

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Abstract. Whether added cloud droplets raise or lower the liquid water path (LWP) of marine stratocumulus is a leading uncertainty in aerosol–cloud radiative forcing. Climate and storm-resolving models represent the competing processes (precipitation suppression, droplet sedimentation and entrainment) very differently depending on their grid spacing. We ran 358 doubly periodic simulations of the drizzling DYCOMS-II RF02 nocturnal stratocumulus with the E3SM atmosphere model EAMxx (SCREAM) for 12 h. The simulations span grid spacings from 40 km to 312 m and droplet number concentrations (N_d) from 20 to 1000 cm^{-3}. They are factorial in a cloud-droplet sedimentation multiplier (0.5, 1, 2) and in the N_d dependence of autoconversion, which we switch on or off. At 10–40 km, where all turbulence is parameterised, LWP increases monotonically with N_d. The high-N_d susceptibility $d\ln\mathrm{LWP}/d\ln N_\mathrm{d}$ is 0.24–0.27, and precipitation suppression accounts for all of it. As the grid is refined, entrainment strengthens and the clouds thin; rain production falls by more than an order of magnitude and most of the rain evaporates before reaching the surface. The precipitation-suppression gain in LWP at $N_\mathrm{d} = 1000\,\mathrm{cm}^{-3}$ shrinks from 134 to 6 $\mathrm{g\,m}^{-2}$. The LWP maximum moves from $N_\mathrm{d} \geq 1000$ to $N_\mathrm{d} \approx 100\,\mathrm{cm}^{-3}$, producing an inverted-V relationship whose high-N_d branch ($S = -0.04$ to -0.05 at 312 m) is entrainment-driven, persists without precipitation suppression and steepens with faster sedimentation. A mixed-layer LWP budget, which reproduces the simulated LWP tendencies (correlation 0.88), attributes the change to an entrainment drying and warming that exceeds radiative and surface forcing at sub-kilometre spacing. Neither LWP nor its N_d response is converged at 312 m, and 3D turbulence has a negligible effect. Global storm-resolving models at kilometre scale therefore sit in a regime where the sign and magnitude of the LWP adjustment are set by the grid rather than by the physics alone.

1 Introduction

Aerosol–cloud interactions remain the largest source of uncertainty in the effective radiative forcing of the climate system (Bellouin et al., 2020; Forster et al., 2021). Adding cloud condensation nuclei to a liquid cloud increases its droplet number concentration N_d and, at fixed liquid water, its albedo (Twomey, 1974, 1977). The cloud then adjusts. Its liquid water path (LWP) and cloud fraction change in response to the altered microphysics, and these adjustments can amplify or offset the Twomey effect (Stevens and Feingold, 2009; Mülmenstädt and Feingold, 2018; Glassmeier et al., 2021).

Two adjustment pathways of opposite sign dominate in marine stratocumulus. Smaller droplets coalesce less efficiently, so drizzle is suppressed and liquid water is retained. This is precipitation suppression, which increases LWP (Albrecht, 1989). Smaller droplets also sediment more slowly and evaporate more readily near cloud top. That enhances entrainment of warm, dry free-tropospheric air, which thins the cloud (Ackerman et al., 2004; Bretherton et al., 2007; Wood, 2007; Hill et al., 2009; Small et al., 2009). Large-eddy simulations (LES) of nocturnal stratocumulus show that the sedimentation–entrainment feedback alone can make LWP decrease with N_d when the overlying air is dry (Ackerman et al., 2004; Bretherton et al., 2007; Uchida et al., 2010). Satellite composites of LWP against N_d have an "inverted-V" shape: LWP increases with N_d in clean, precipitating clouds and decreases in polluted, non-precipitating ones (Gryspeerdt et al., 2019; Possner et al., 2020). Pollution tracks show LWP responses of both signs that are weak on average (Toll et al., 2019). LES ensembles reproduce this shape (Glassmeier et al., 2019, 2021; Hoffmann et al., 2020). LWP decreases may therefore offset a substantial part of the Twomey effect (Toll et al., 2019; Glassmeier et al., 2021). This interpretation is debated, because the observed relationship can be confounded by meteorological covariability, by retrieval error and by the slow, transient response of the cloud (Gryspeerdt et al., 2019; Arola et al., 2022; Glassmeier et al., 2021; Zhang et al., 2022; Feingold et al., 2022).

Global climate models (GCMs) generally produce positive LWP responses to aerosol (Quaas et al., 2009; Michibata et al., 2016; Gettelman, 2015; Mülmenstädt et al., 2020). Even so, present-day spatial LWP–N_d correlations in GCMs can be negative (Mülmenstädt et al., 2024b), and the parameterisations GCMs use to represent cloud-top entrainment may not reproduce adjustments seen in process models (Mülmenstädt et al., 2024a; Karset et al., 2020). Global storm-resolving models (GSRMs), with grid spacings of a few kilometres, are now run routinely (Stevens et al., 2019; Caldwell et al., 2021; Donahue et al., 2024). Some studies suggest that resolving cloud-scale circulations improves the simulated aerosol effect on cloud water (Sato et al., 2018; Terai et al., 2020). However, kilometre-scale grids lie in the turbulence "grey zone" (or "terra incognita") for the stratocumulus-topped boundary layer. There, the energy-containing eddies, a few hundred metres across, are neither resolved nor fully parameterised (Wyngaard, 2004; Honnert et al., 2020; de Roode et al., 2019). Even LES of stratocumulus with grid spacings of tens of metres are sensitive to resolution and numerics, mostly through entrainment (Bretherton et al., 1999; Stevens et al., 2005; Pressel et al., 2017; Mellado et al., 2018). Doubly periodic configurations of the Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM) show that cloud structure in stratocumulus cases changes systematically between 5 km and a few hundred metres (Bogenschutz et al., 2023a, b).

How the balance of adjustment pathways changes across this range of grid spacings has not been isolated. The answer matters for interpreting aerosol forcing in GSRMs and for using GSRM or LES output to constrain GCM parameterisations. Here we address four questions using a factorial set of doubly periodic simulations of the drizzling DYCOMS-II RF02 stratocumulus with EAMxx:

1. How does the LWP susceptibility to N_d depend on horizontal grid spacing from 40 km to 312 m?
2. Which microphysical pathway (precipitation suppression, or sedimentation-mediated entrainment) controls it at each spacing?
3. What boundary-layer processes cause the change?

4. What does the answer imply for GSRMs, GCMs and satellite-based estimates?

The experiment switches the N_d dependence of autoconversion on and off and scales the cloud-droplet fall speed. This lets us separate the two pathways directly, in the spirit of mechanism-denial experiments.

2 Model, case and methods

2.1 DP-EAMxx

EAMxx is the C++/Kokkos atmosphere model of the Energy Exascale Earth System Model (E3SM), also known as SCREAM (Caldwell et al., 2021; Donahue et al., 2024). It solves the nonhydrostatic equations with a spectral-element dynamical core on fourth-order elements (Taylor et al., 2020). Physics is computed on a separate finite-volume grid with 2×2 columns per element ("pg2"; Hannah et al., 2021). Turbulence, cloud macrophysics and condensation are handled by the Simplified Higher-Order Closure (SHOC; Bogenschutz and Krueger, 2013). SHOC is an assumed-PDF scheme that diagnoses cloud fraction and liquid water from prognostic turbulent kinetic energy and second- and third-order moments. Microphysics uses the Predicted Particle Properties scheme (P3; Morrison and Milbrandt, 2015). In P3, warm rain forms by the Khairoutdinov and Kogan (2000) autoconversion and accretion rates. P3 also sediments cloud droplets with a gamma size distribution, using mass- and number-weighted fall speeds. Radiation is computed by RRTMGP (Pincus et al., 2019) every third physics step. Droplet activation uses a prescribed CCN number concentration (the "SPA" option), scaled to set the target N_d. In this configuration, condensation and evaporation of cloud water are part of SHOC's saturation adjustment and do not depend on droplet size; P3's vapour–liquid exchange consists only of rain evaporation, which we confirmed in the model output. The droplet-size dependence of LWP therefore enters only through autoconversion, cloud-droplet sedimentation and the effective radius used in radiation.

We use the doubly periodic configuration (DP-EAMxx; Bogenschutz et al., 2023a). It applies large-scale forcing from an intensive-observation-period (IOP) file to a horizontally periodic domain. All simulations use 128 vertical levels, with a grid spacing of about 50 m near the inversion. An optional horizontal ("3D") turbulence closure couples SHOC's eddy diffusivity to the dynamical core. It is off in the main experiment and is tested at 625 m and 312 m (Appendix C).

Two runtime options were added to P3 for this study:

1. A `cloud_sedimentation_factor` s, which multiplies the mass- and number-weighted cloud-droplet fall speeds. It is analogous to the `cld_sed` parameter of the MG2 scheme (Morrison and Gettelman, 2008; Gettelman and Morrison, 2015); $s = 1$ is the default.

2. Configurable prefactor and N_d exponent of the KK2000 autoconversion rate,

$$\left(\frac{\partial q_\mathrm{r}}{\partial t}\right)_\mathrm{auto} = A\, q_\mathrm{c}^{2.47} \left(\frac{N_\mathrm{d}}{1\,\mathrm{cm}^{-3}}\right)^{-\beta}, \tag{1}$$

   where q_c is the cloud water mixing ratio.

The default KK2000 values, $A = 1350$ and $\beta = 1.79$, are the "precipitation suppression on" configuration. In the "off" configuration, $\beta = 0$ and $A = 1350 \times 63^{-1.79} = 0.812$, so the two rates are equal at $N_\mathrm{d} = 63\,\mathrm{cm}^{-3}$, the geometric centre of the N_d

range. Accretion in KK2000 does not depend on N_{d}. Switching suppression off therefore removes the direct N_{d} control on rain formation, but leaves rain itself, and its dependence on cloud water, intact.

2.2 Case

The DYCOMS-II RF02 case is a nocturnal, drizzling stratocumulus deck over the north-eastern Pacific, about 800 m deep (Stevens et al., 2003; vanZanten et al., 2005). It was the subject of an LES intercomparison (Ackerman et al., 2009). We use the E3SM single-column IOP file for this case. The forcing is constant over 12 h:

- Surface fluxes are prescribed: sensible 16 W m^{-2} and latent 93 W m^{-2}.
- Large-scale subsidence is applied as a forcing tendency.
- Temperature and moisture are nudged towards the initial profiles above 650 hPa with a 1 h time scale.
- Shortwave radiation is off; winds are not nudged and there is no Coriolis force.

The initial boundary layer has $\theta_l \approx 288.3$ K and $q_t \approx 9.45\,\mathrm{g\,kg^{-1}}$, capped by an inversion at about 795 m. All runs last 12 h, and unless stated otherwise we analyse hours 10–12.

2.3 Experiment design

Table 1 lists the factors. The full matrix crosses the following:

- eight grid spacings (40, 20, 10, 5, 2.5 and 1.25 km, 625 and 312 m);
- six target droplet numbers (20, 50, 100, 200, 500 and 1000 $\mathrm{cm^{-3}}$);
- three sedimentation factors;
- precipitation suppression on or off.

That makes 288 simulations with 3D turbulence off. A further 72 simulations repeat the 625 m and 312 m grids with 3D turbulence on. Of these, 70 were complete at the time of writing; two 312 m runs at $N_{\mathrm{d}} = 1000\,\mathrm{cm^{-3}}$ were still running, so we analyse 358 runs in all. Hereafter we refer to grids by their nominal spacing Δx, the mean spacing of the dynamical-core quadrature points.

Table 2 gives the numerical settings. Physics columns are spaced $1.5\Delta x$ apart. The domain is 240 km wide for $\Delta x \le 10$ km. It is 600 km at 40 km and 300 km at 20 km, so that each coarse domain still holds 10×10 physics columns. At $\Delta x \ge 10$ km there are 100–256 columns, no turbulence is resolved, and the runs behave as an ensemble of nearly identical single columns. The physics time step is 100 s for $\Delta x \ge 5$ km and 75, 60, 30 and 15 s at 2.5 km, 1.25 km, 625 m and 312 m, respectively. The 625 m and 312 m runs were integrated in several restart segments. We verified that segmented and continuous integrations are bit-for-bit identical. An earlier 6 h set of 216 runs (40 km to 1.25 km) reproduces the first 6 h of the present runs bit-for-bit.

Table 1. Experiment factors. Every combination is run for 12 h with 3D turbulence off (288 runs). The 625 m and 312 m grids are repeated with 3D turbulence on (72 runs, 70 analysed).

Factor	Levels	Implementation
Grid spacing Δx	40, 20, 10, 5, 2.5, 1.25 km; 625, 312 m	Table 2
Target N_{d}	20, 50, 100, 200, 500, 1000 cm^{-3}	SPA CCN-to-N_{d} factor (exponent 1)
Sedimentation factor s	0.5, 1, 2	multiplies cloud-droplet fall speeds in P3
Precipitation suppression	on / off	Eq. (1): $(A, \beta) = (1350, 1.79)$ / $(0.812, 0)$
3D turbulence	off; on at 625 and 312 m	SHOC eddy diffusivity applied horizontally

Table 2. Numerical settings. Δx is the nominal (dynamics) spacing; physics columns are $1.5\Delta x$ apart. N_{col} is the number of physics columns, Δt_{phys} the physics time step, Δt_{dyn} the dynamics step and ν_{top} the sponge-layer diffusivity.

Δx	domain (km)	elements	N_{col}	Δt_{phys} (s)	Δt_{dyn} (s)	ν_{top} ($\mathrm{m}^2\,\mathrm{s}^{-1}$)
40 km	600	5×5	100	100	8.33	1.2×10^5
20 km	300	5×5	100	100	8.33	6.0×10^4
10 km	240	8×8	256	100	8.33	3.0×10^4
5 km	240	16×16	1024	100	8.33	1.5×10^4
2.5 km	240	32×32	4096	75	6.25	7500
1.25 km	240	64×64	16 384	60	2.50	1667
625 m	240	128×128	65 536	30	1.25	833
312 m	240	256×256	262 144	15	0.625	417

All runs: 128 levels; radiation every third physics step; one macrophysics–microphysics subcycle per physics step; rain fraction set to one.

Droplet number is set through the SPA CCN scaling factor. Because P3 depletes and replenishes N_{d} through collision–coalescence, evaporation and activation, the diagnosed in-cloud N_{d} is not exactly the target. It is 21–64 % above target at $\Delta x \geq 10$ km and between 21 % below and 11 % above target at 312 m. All susceptibilities below are therefore computed with the diagnosed in-cloud N_{d}: the median over cloudy grid cells (cloud water $> 10^{-5}\,\mathrm{kg\,kg}^{-1}$) of the 3D snapshots at 10 and 12 h.

2.4 Diagnostics

The diagnostics come from four output streams:

- domain-mean profiles averaged over 15 min, including SHOC moments and P3 process rates;
- 15 min instantaneous column fields (LWP, surface rain);

– 15 min cloud-top fields from an AeroCom-style operator (cloud-top liquid fraction and fraction-weighted droplet number and effective radius);

– 2-hourly 3D snapshots.

2.4.1 Bulk and cloud-structure quantities.

LWP is the domain mean. Cloud cover is the fraction of columns with LWP $> 10\,\mathrm{g\,m^{-2}}$. The inversion height z_i is where the domain-mean liquid potential temperature θ_l crosses the midpoint between its boundary-layer (below 400 m) and free-tropospheric (1400 m) values. Cloud base z_b is the lowest level at which the domain-mean liquid cloud fraction reaches half its maximum. The entrainment rate is

$$w_\mathrm{e} = \frac{dz_\mathrm{i}}{dt} - w_\mathrm{s}(z_\mathrm{i}), \tag{2}$$

where dz_i/dt is a linear fit over hours 8–12 and w_s is the IOP subsidence velocity at the mean z_i. A 4 h fit is less sensitive than a 2 h fit to the 50 m level spacing at the inversion; values for hours 10–12 alone are within about $1\,\mathrm{mm\,s^{-1}}$.

2.4.2 Turbulence and boundary-layer structure.

The resolved vertical-velocity variance and skewness are computed from the 3D snapshots at 10 and 12 h, with $w = -\omega/(\rho g)$. They are compared with SHOC's subgrid variance averaged over the boundary layer. Decoupling is measured following Jones et al. (2011), as the difference in q_t (Δq_t) and θ_l between the lowest and highest quarters of the boundary layer. The dominant cell scale λ is the wavelength of the peak of the radially averaged LWP power spectrum, as in Wood and Hartmann (2006).

2.4.3 Susceptibility.

We define the LWP susceptibility between two droplet-number levels a and b, run otherwise identically, as

$$S_{a\to b} = \frac{\ln \mathrm{LWP}_b - \ln \mathrm{LWP}_a}{\ln N_{\mathrm{d},b} - \ln N_{\mathrm{d},a}}, \tag{3}$$

with LWP averaged over hours 10–12 and in-cloud N_d. We report three branches: $S_{20\to50}$ (clean), $S_{100\to200}$ (moderate) and $S_{200\to1000}$ (polluted). As a measure of sampling uncertainty, we recompute S for each of the eight 15 min intervals in hours 10–12 and quote the standard deviation. This does not capture the spread across initial-condition ensembles, which we did not run (Sect. 4.4).

2.5 Factorial decomposition

The difference between otherwise identical runs with precipitation suppression on and off isolates the precipitation-suppression contribution,

$$S_\mathrm{ps}(s) = S_\mathrm{on}(s) - S_\mathrm{off}(s). \tag{4}$$

With suppression off, the remaining $N_{\rm d}$ dependence of LWP comes from droplet sedimentation, from the effective radius in radiation, and from indirect interactions. The sedimentation flux scales linearly with s, so to first order the sedimentation contribution to S is proportional to s. We therefore fit

$$S_{\rm off}(s) = a + b\,s \tag{5}$$

over $s = 0.5$, 1 and 2, and write

$$S_{\rm on}(s) = S_{\rm ps}(s) + b\,s + a. \tag{6}$$

Here $b\,s$ is the sedimentation contribution and a is a residual: the $S_{\rm off}$ extrapolated to zero sedimentation. The residual includes radiative effective-radius effects and any non-linearity of the response in s; root-mean-square deviations of the linear fit are below 0.02.

2.6 Mixed-layer LWP budget

To link the $N_{\rm d}$ response to boundary-layer processes, we evaluate a bulk budget of the domain-mean boundary layer for every run in 2 h windows. It follows mixed-layer theory (Lilly, 1968; Stevens, 2002) and the LWP-tendency framework of van der Dussen et al. (2014). For a well-mixed layer of depth $z_{\rm i}$ with conserved variables θ_l and q_t,

$$\frac{d\theta_l}{dt} = \frac{1}{z_{\rm i}}\left[\frac{H}{\rho c_p} + w_{\rm e}\,\Delta\theta_l - \frac{F_{\rm rad}}{\rho c_p} + \frac{P}{\rho c_p}\right], \tag{7}$$

$$\frac{dq_t}{dt} = \frac{1}{z_{\rm i}}\left[\frac{E}{\rho L_v} + w_{\rm e}\,\Delta q_t - \frac{P}{\rho L_v}\right], \tag{8}$$

$$\frac{dz_{\rm i}}{dt} = w_{\rm e} + w_{\rm s}(z_{\rm i}). \tag{9}$$

The terms are:

- H and E, the prescribed sensible and latent heat fluxes;
- $F_{\rm rad}$, the radiative cooling of the layer below $z_{\rm i} + 150$ m;
- P, the surface rain rate in energy units;
- $\Delta\theta_l$ and Δq_t, the jumps between the mixed layer (0.1–0.8 $z_{\rm i}$) and 100 m above $z_{\rm i}$.

Cloud base follows the lifting condensation level of mixed-layer air,

$$\frac{dz_{\rm b}}{dt} = \frac{\partial z_{\rm b}}{\partial \theta_l}\frac{d\theta_l}{dt} + \frac{\partial z_{\rm b}}{\partial q_t}\frac{dq_t}{dt}, \tag{10}$$

with sensitivities computed numerically (120–130 m K^{-1} and -190 to -210 m (g kg^{-1})$^{-1}$). Saturation follows Bolton (1980). Assuming LWP $\propto h^2$ with cloud thickness $h = z_{\rm i} - z_{\rm b}$ and fixed adiabaticity,

$$\frac{d\ln \mathrm{LWP}}{dt} = \frac{2}{h}\left(\frac{dz_{\rm i}}{dt} - \frac{dz_{\rm b}}{dt}\right). \tag{11}$$

Equation (11) is linear in the forcing terms, so it separates into contributions from entrainment (w_e in dz_i/dt and in Eqs. 7–8), subsidence, radiation, surface fluxes and precipitation. The budget reproduces the simulated tendencies over all 1440 run-windows with 3D turbulence off. The correlations are 0.89 for $d\theta_l/dt$, 0.87 for dq_t/dt and 0.88 for $d\mathrm{LWP}/dt$, and the root-mean-square error in $d\mathrm{LWP}/dt$ is $4.2\,\mathrm{g\,m^{-2}\,h^{-1}}$, against a standard deviation of $8.4\,\mathrm{g\,m^{-2}\,h^{-1}}$ (Appendix B).

3 Results

3.1 Cloud structure across grid spacings

Figure 1 shows LWP at hour 12 for the default sedimentation factor with precipitation suppression on. Three changes stand out as the grid is refined.

- The cells shrink, from about 30 km at 2.5 km spacing to 13–20 km at 625 and 312 m.
- The mean LWP falls; at a target N_d of $200\,\mathrm{cm^{-3}}$ it decreases from $214\,\mathrm{g\,m^{-2}}$ at 5 km to $74\,\mathrm{g\,m^{-2}}$ at 312 m.
- Clear gaps open; cloud cover at the same N_d falls from 1.00 to 0.84.

At $N_\mathrm{d} = 20\,\mathrm{cm^{-3}}$ the sub-kilometre clouds break into small, bright drizzling cells, and cloud cover drops to 0.50 at 312 m.

The mean state changes in the same direction. As Δx decreases from 40 km to 312 m (target $N_\mathrm{d} = 200\,\mathrm{cm^{-3}}$, suppression on):

- the inversion rises from 863 to 1016 m;
- cloud base rises from 195 to 720 m, so the cloud layer thins from 668 to 296 m;
- the mixed layer becomes 1.8 K warmer.

The 40 and 20 km runs are nearly indistinguishable, and the 10 km runs differ from them only at high N_d (Table 3). Resolved motions carry no variance at these spacings (Sect. 3.7). They are the "single-column" limit of the model, analogous to a GCM column using the same physics.

3.2 The LWP–N_d relationship and its dependence on grid spacing

Figure 2 and Table 3 show the central result. With precipitation suppression on, LWP increases monotonically with N_d at $\Delta x \geq 10$ km, from 135 to $342\,\mathrm{g\,m^{-2}}$ at 40 km. As the grid is refined, the curve flattens at high N_d and then turns over. The LWP maximum (stars in Fig. 2a) moves from $N_\mathrm{d} \geq 1000\,\mathrm{cm^{-3}}$ at $\Delta x \geq 10$ km to a target of $500\,\mathrm{cm^{-3}}$ at 5 km, $200\,\mathrm{cm^{-3}}$ from 2.5 km to 625 m, and $100\,\mathrm{cm^{-3}}$ (in-cloud $84\,\mathrm{cm^{-3}}$) at 312 m. The resulting inverted-V shape matches the shape seen in satellite composites and LES ensembles (Gryspeerdt et al., 2019; Glassmeier et al., 2021). In these simulations, however, it emerges only once the boundary-layer eddies begin to be resolved.

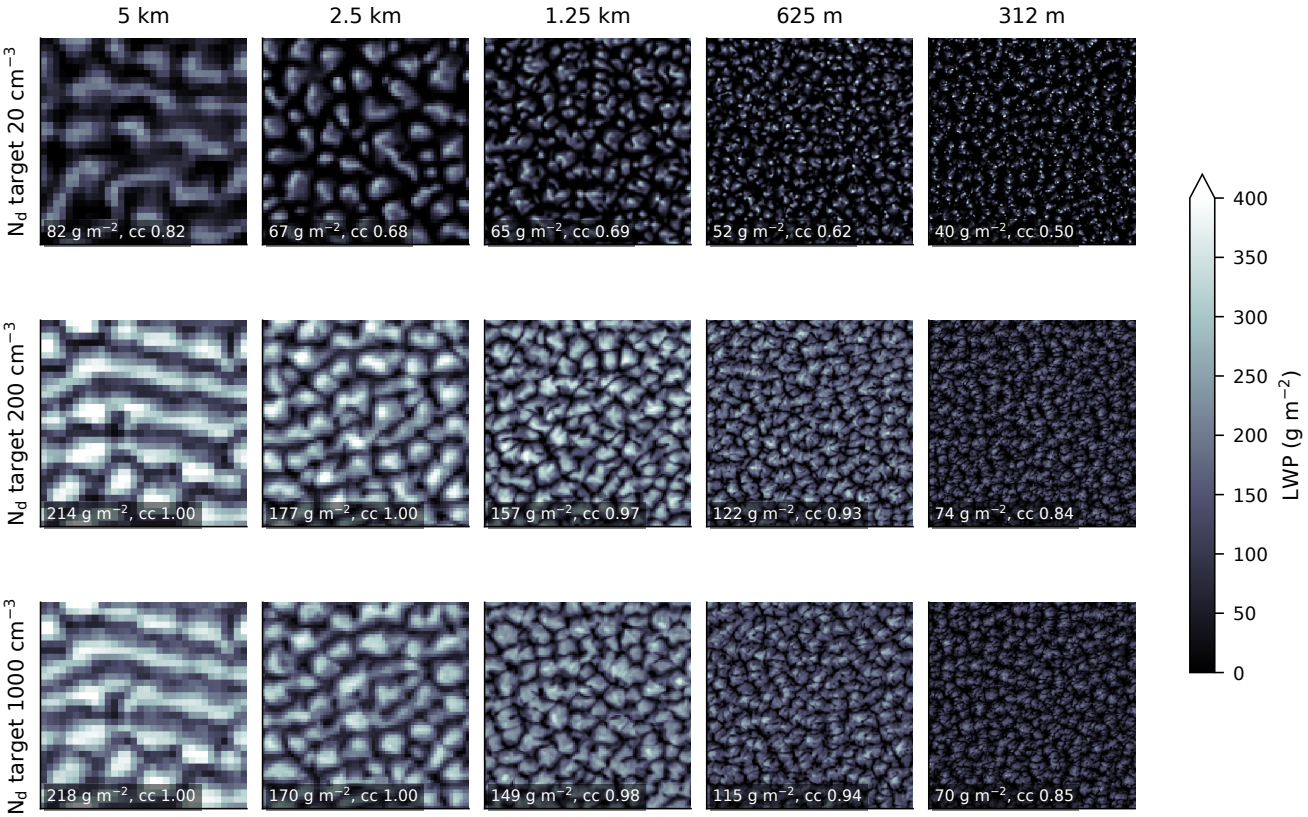


Figure 1. LWP at hour 12 over the full 240 km domain for target N_d of 20, 200 and 1000 cm^{-3} (rows) and grid spacings from 5 km to 312 m (columns). Sedimentation factor 1, precipitation suppression on, 3D turbulence off. Labels give the domain-mean LWP and cloud cover averaged over hours 10–12.

With precipitation suppression off (Fig. 2b), LWP is nearly independent of N_d at every spacing. It decreases slightly with N_d, except at 5 and 2.5 km where it first increases. Surface precipitation (Fig. 2d) falls steeply with N_d when suppression is on, and falls faster at finer spacing. At 312 m it is below 0.01 mm d^{-1} for $N_\mathrm{d} \geq 100\,\mathrm{cm}^{-3}$. Entrainment (Fig. 2c) roughly doubles from the coarse to the fine grids. It also increases with N_d at $\Delta x \leq 5$ km whether suppression is on or off, but hardly at all at $\Delta x \geq 20$ km.

Figure 3 and Table 4 quantify the response as susceptibilities. With suppression on and $s = 1$, the moderate-N_d susceptibility $S_{100\to200}$ decreases monotonically with Δx: 0.27 at 40 km, 0.20 at 5 km, 0.11 at 2.5 km, 0.07 at 1.25 km, 0.03 at 625 m and -0.05 at 312 m. The polluted branch $S_{200\to1000}$ changes sign between 5 km (0.01) and 2.5 km (-0.03), and stays between -0.03 and -0.04 down to 312 m. The clean branch behaves differently. $S_{20\to50}$ increases from 0.39 at $\Delta x \geq 10$ km to 0.65–0.76 at $\Delta x \leq 2.5$ km and then changes little with further refinement.

Table 3. Domain-mean LWP ($\mathrm{g\,m^{-2}}$, hours 10–12) for sedimentation factor 1 and 3D turbulence off. Columns are target N_d ($\mathrm{cm^{-3}}$). Bold marks the maximum with precipitation suppression on.

Δx	supp.	20	50	100	200	500	1000
40 km	on	135	188	246	295	333	**342**
	off	234	221	216	211	209	208
20 km	on	135	187	244	292	328	**336**
	off	233	220	215	211	209	207
10 km	on	134	186	240	281	308	**312**
	off	232	217	213	208	206	204
5 km	on	82	137	184	214	**222**	218
	off	135	149	153	158	159	160
2.5 km	on	67	119	163	**177**	175	170
	off	117	137	139	138	137	136
1.25 km	on	65	111	149	**157**	154	149
	off	116	127	127	124	122	122
625 m	on	52	98	120	**122**	119	115
	off	103	109	108	103	101	99
312 m	on	40	72	**76**	74	72	70
	off	79	79	72	69	66	64

The sampling uncertainty is small at sub-kilometre spacing (standard deviation ≤ 0.03). It is larger at $\Delta x \geq 5$ km (up to 0.1 for $S_{100\to200}$), where LWP oscillates on 15 min time scales. The sedimentation factor matters most where entrainment dominates. At $\Delta x \leq 2.5$ km with suppression on, $S_{200\to1000}$ is -0.010 to -0.016 for $s = 0.5$ and -0.063 to -0.086 for $s = 2$ (Fig. 3c). With suppression off, S is within ±0.13 of zero at $\Delta x \geq 10$ km on all three branches, and negative on the two high-N_d branches at $\Delta x \leq 1.25$ km for $s \leq 1$. At 312 m, $S_{100\to200}$ is -0.07 and $S_{200\to1000}$ is -0.05 for $s = 1$.

3.3 Transient development of the inverted V

The relationship develops over the whole 12 h (Fig. 4).

- At $\Delta x \geq 10$ km, $S_{100\to200}$ grows steadily, from 0.12 in hours 2–4 to 0.27 in hours 10–12. High-N_d clouds keep accumulating the water that low-N_d clouds rain out (Fig. 4d); at 40 km and high N_d, LWP is still rising by up to $15\,\mathrm{g\,m^{-2}\,h^{-1}}$ in hours 10–12.

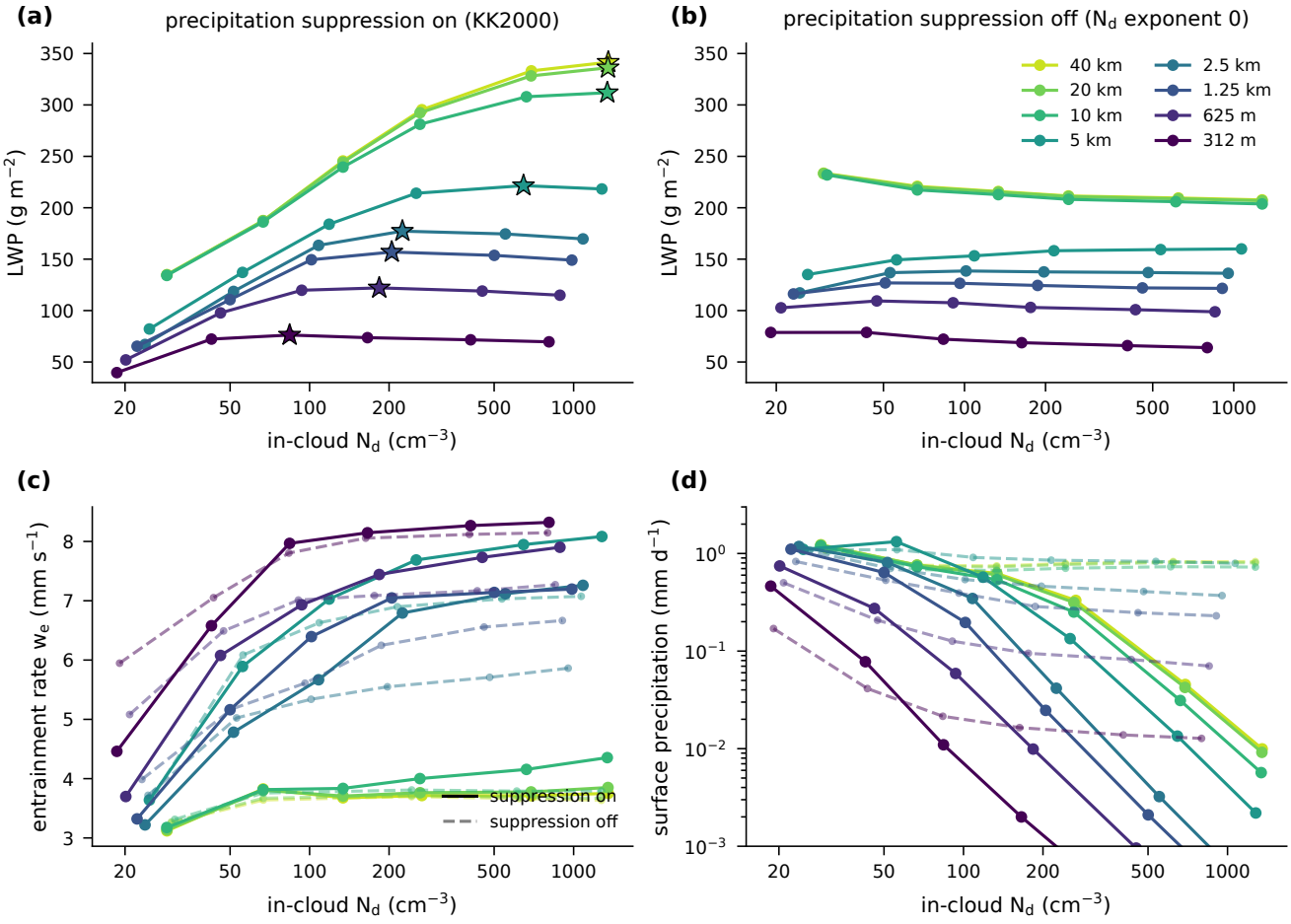


Figure 2. Response to N_d at every grid spacing (colours) for sedimentation factor 1. (a) LWP with precipitation suppression on; stars mark the maximum. (b) LWP with suppression off. (c) Entrainment rate (8–12 h fit) and (d) surface precipitation, with suppression on (solid) and off (dashed). Values are hour 10–12 means plotted against the diagnosed in-cloud N_d.

- At $\Delta x \leq 1.25$ km, $S_{100\to200}$ and $S_{200\to1000}$ start near zero and drift negative. At 312 m, $S_{100\to200}$ goes from 0.004 in hours 2–4 to -0.05 in hours 10–12, and $S_{200\to1000}$ from -0.006 to -0.036.
- $S_{20\to50}$ is large from the start at sub-kilometre spacing (0.51–0.63 in hours 2–4) and grows slowly.

At $\Delta x \leq 2.5$ km, LWP decreases through most of the simulation (Fig. 4e,f). In hours 10–12 it is still decreasing in 98 % of these runs, by at most 4.4 g m^{-2} h^{-1} (up to 6 % h^{-1}). These simulations are therefore not at steady state, and the high-N_d susceptibilities at fine spacing were still becoming more negative at the end of the runs. This is consistent with the long adjustment time scales found in LES (Glassmeier et al., 2021; Hoffmann et al., 2020).

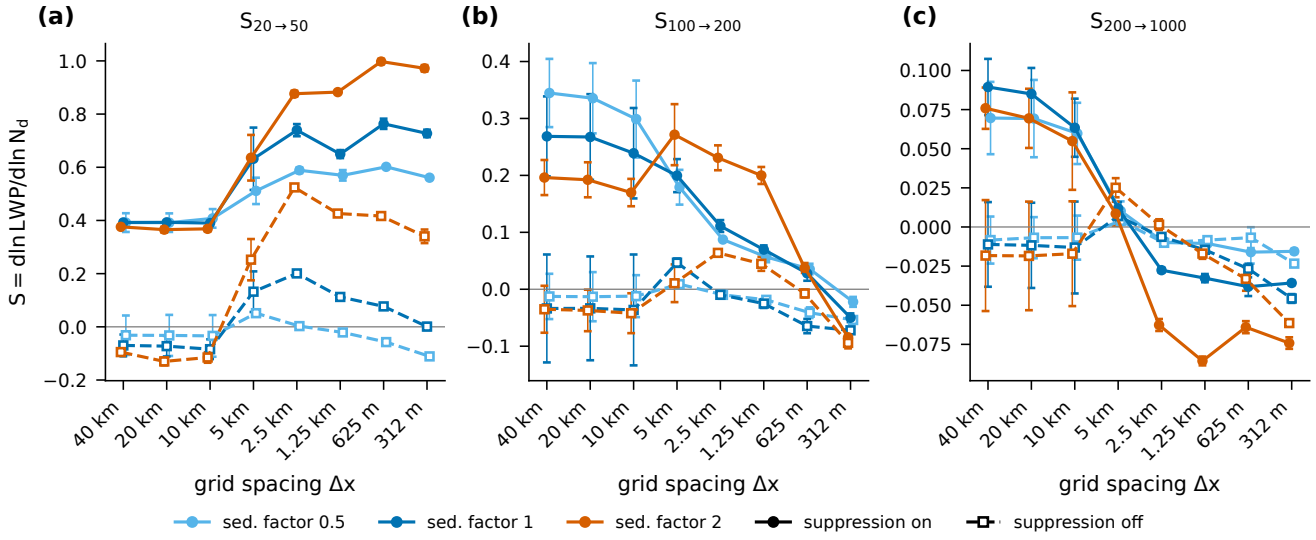


Figure 3. LWP susceptibility on the (a) clean, (b) moderate and (c) polluted N_d branches against grid spacing. Colours: sedimentation factor. Solid lines with filled circles: precipitation suppression on. Dashed lines with open squares: off. Error bars: standard deviation of S over the eight 15 min intervals in hours 10–12. Symbols are offset horizontally for clarity.

Table 4. Susceptibility $S = d\ln\mathrm{LWP}/d\ln N_\mathrm{d}$ on three branches for sedimentation factor 1 and 3D turbulence off. In parentheses: standard deviation over the eight 15 min intervals in hours 10–12. The last column is the target N_d (cm^{-3}) of maximum LWP with suppression on.

	$S_{20\to50}$		$S_{100\to200}$		$S_{200\to1000}$		N_d at
Δx	on	off	on	off	on	off	max LWP
40 km	0.393 (0.007)	−0.070 (0.041)	0.269 (0.070)	−0.034 (0.095)	0.089 (0.018)	−0.011 (0.027)	1000
20 km	0.393 (0.006)	−0.073 (0.039)	0.267 (0.075)	−0.034 (0.091)	0.085 (0.017)	−0.012 (0.027)	1000
10 km	0.390 (0.011)	−0.084 (0.037)	0.239 (0.079)	−0.036 (0.097)	0.063 (0.019)	−0.013 (0.029)	1000
5 km	0.632 (0.117)	0.132 (0.076)	0.200 (0.029)	0.047 (0.007)	0.012 (0.004)	0.007 (0.002)	500
2.5 km	0.740 (0.023)	0.201 (0.010)	0.111 (0.010)	−0.010 (0.005)	−0.028 (0.002)	−0.006 (0.001)	200
1.25 km	0.650 (0.015)	0.112 (0.004)	0.070 (0.007)	−0.025 (0.008)	−0.033 (0.003)	−0.015 (0.001)	200
625 m	0.764 (0.019)	0.077 (0.003)	0.028 (0.013)	−0.065 (0.013)	−0.038 (0.006)	−0.027 (0.003)	200
312 m	0.727 (0.014)	0.001 (0.013)	−0.050 (0.008)	−0.072 (0.006)	−0.036 (0.001)	−0.046 (0.003)	100

3.4 Attribution to precipitation suppression and sedimentation

Figure 5 shows the factorial decomposition of Eq. (6) for $s = 1$.

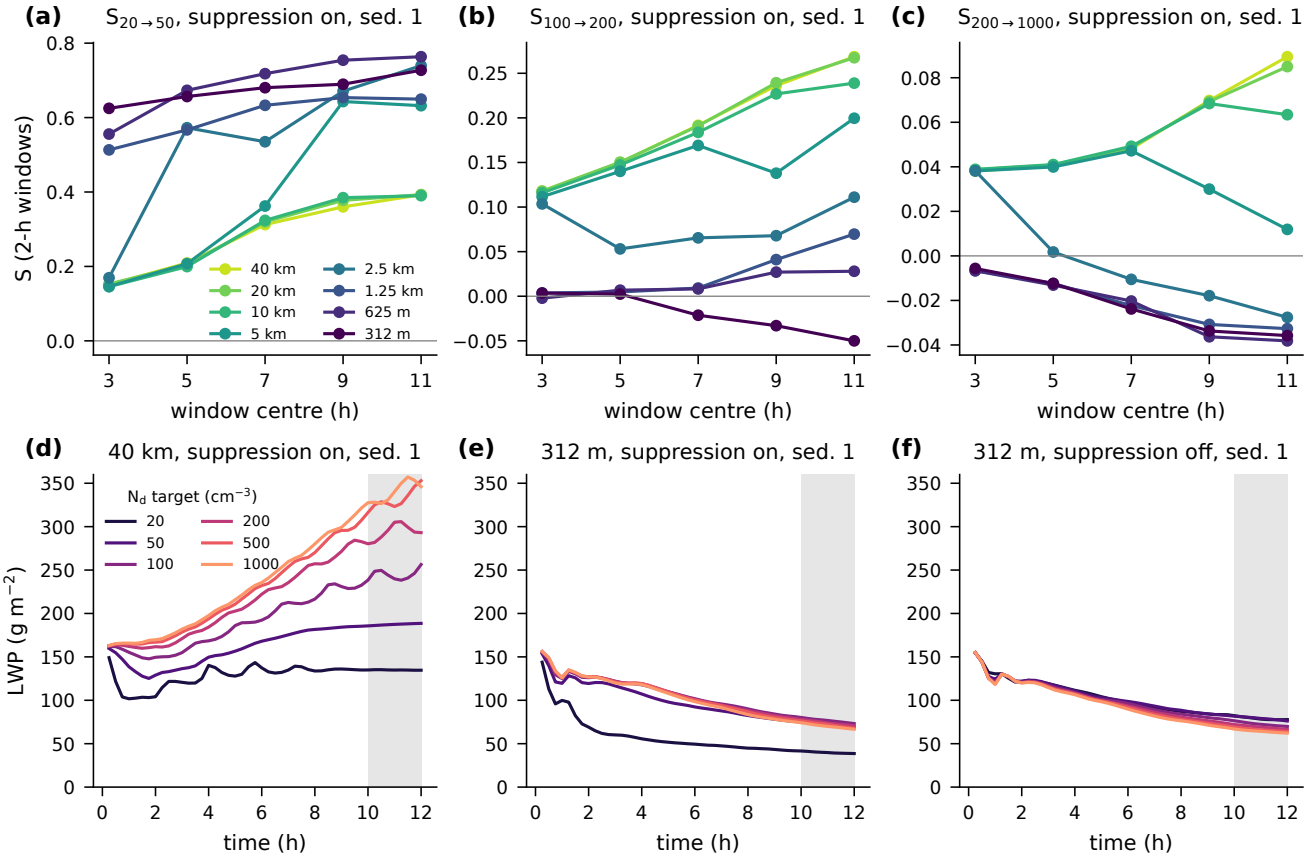


Figure 4. (a–c) Susceptibility on the three branches for consecutive 2 h windows (suppression on, sedimentation factor 1), coloured by grid spacing. (d–f) LWP time series for each target N_{d} at 40 km with suppression on, and at 312 m with suppression on and off. Grey shading marks the analysis window.

3.4.1 Moderate and polluted branches.

At $\Delta x \geq 10$ km, precipitation suppression accounts for the whole of $S_{100\to200}$ and $S_{200\to1000}$: $S_{\mathrm{ps}} = 0.28$–0.30 and 0.08–0.10. The sedimentation and residual terms are each between -0.02 and -0.003. As the grid is refined, S_{ps} on the moderate branch falls to 0.12 at 2.5 km, 0.09 at 1.25 km and 625 m, and 0.02 at 312 m. On the polluted branch it is indistinguishable from zero (-0.02 to $+0.01$) at $\Delta x \leq 2.5$ km. The negative $S_{200\to1000}$ at fine spacing comes from the sedimentation term (-0.016 and -0.024 at 625 m and 312 m) and from the residual (-0.004 to -0.016). Faster droplet sedimentation thus raises LWP more in clean (large-droplet) clouds than in polluted ones, so the entrainment-driven falling branch steepens with s (Fig. 3c), as proposed by Bretherton et al. (2007). With suppression on, dS/ds on the polluted branch is -0.03 to -0.05 at $\Delta x \leq 2.5$ km.

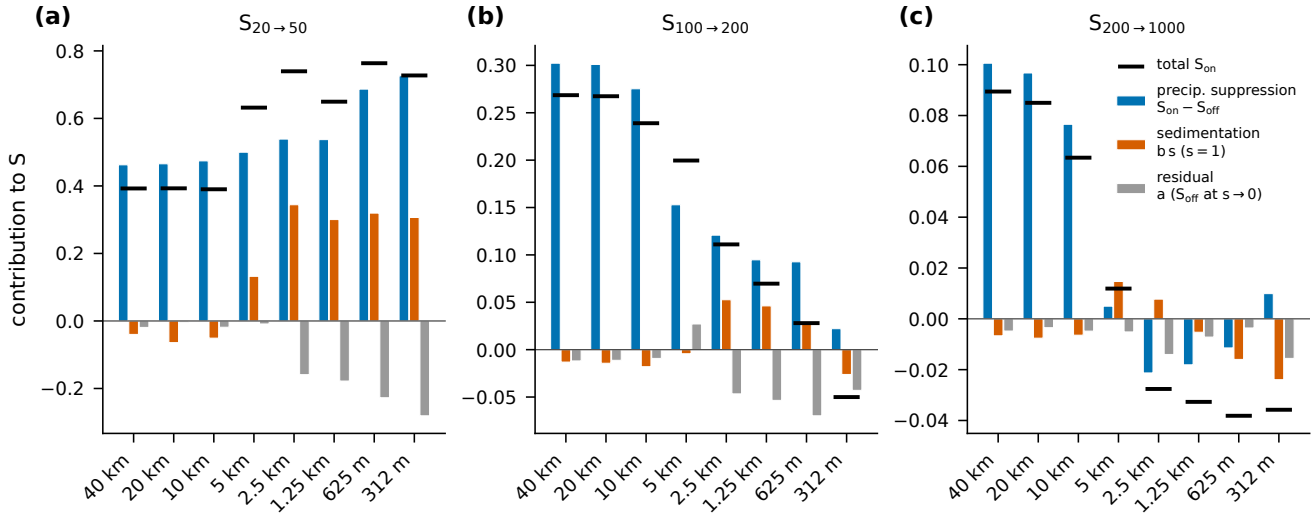


Figure 5. Factorial decomposition (Eq. 6) of the susceptibility with suppression on and sedimentation factor 1 (black bars), for the (a) clean, (b) moderate and (c) polluted branches. Blue: precipitation suppression ($S_{\mathrm{on}} - S_{\mathrm{off}}$). Orange: sedimentation ($b\,s$). Grey: residual (S_{off} extrapolated to zero sedimentation).

3.4.2 Clean branch.

Here the sedimentation term is positive at $\Delta x \leq 5\,\mathrm{km}$ (0.13 at 5 km and about 0.30–0.34 at $\leq 2.5\,\mathrm{km}$). At $N_{\mathrm{d}} \approx 20\,\mathrm{cm}^{-3}$ the droplets are large: cloud-top effective radii are 10–17 μm at $\Delta x \leq 5\,\mathrm{km}$, with Stokes fall speeds of about 1–3.5 $\mathrm{cm\,s}^{-1}$. Droplet sedimentation then acts as a water sink akin to drizzle rather than as a brake on entrainment, and stronger sedimentation lowers LWP most at the lowest N_{d}. The clean-branch residual is large and negative at $\Delta x \leq 2.5\,\mathrm{km}$ (-0.16 to -0.28), compensating part of the larger S_{ps} (0.54–0.73). Because the residual comes from extrapolation in s, its magnitude is uncertain wherever S_{off} is curved in s. This is most evident on the clean branch at sub-kilometre spacing ($S_{\mathrm{off}} = -0.11$, 0.00, 0.34 for $s = 0.5$, 1, 2 at 312 m). We interpret it only as evidence that pathways other than linear sedimentation, including radiative effective-radius effects on thin, broken clouds, contribute there.

3.4.3 Cloud cover.

Most of the clean-branch increase at fine spacing is an increase in cloud cover. Decomposing $d\ln\mathrm{LWP} = d\ln f_{\mathrm{c}} + d\ln\mathrm{LWP}_{\mathrm{ic}}$, where f_{c} is cloud cover and $\mathrm{LWP}_{\mathrm{ic}}$ the mean LWP of cloudy columns, the cloud-cover part of $S_{20\to50}$ at 312 m is 0.53 of 0.73. On the high-N_{d} branches, cloud cover contributes less than 0.03.

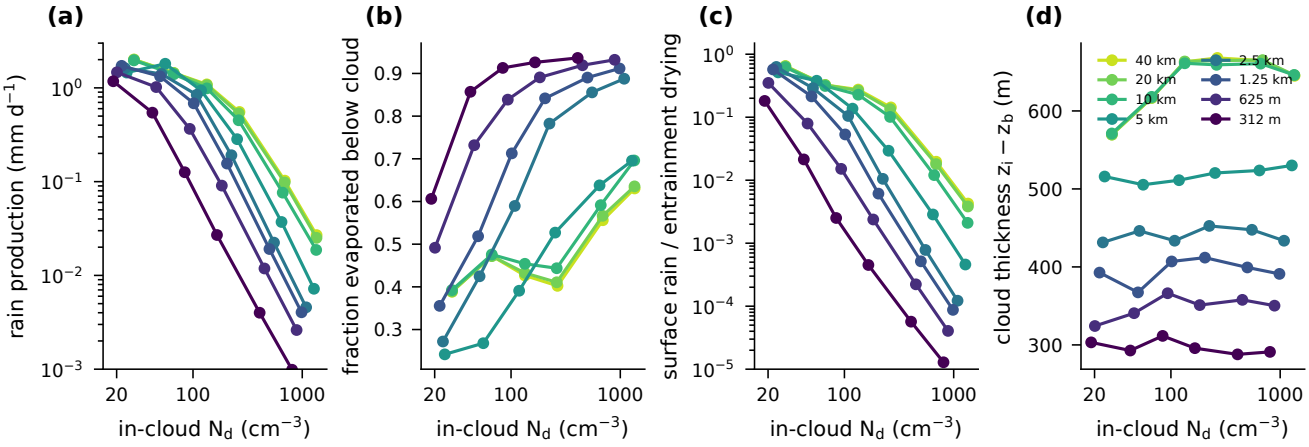


Figure 6. Precipitation budget against in-cloud N_d for every grid spacing (suppression on, sedimentation factor 1). (a) Column rain production (autoconversion plus accretion). (b) Fraction of the production that evaporates before reaching the surface. (c) Ratio of surface rain to the entrainment drying flux. (d) Cloud thickness $z_\mathrm{i} - z_\mathrm{b}$.

3.5 Why precipitation suppression loses its leverage

Precipitation suppression can raise LWP only to the extent that precipitation is a significant sink of boundary-layer water. Figure 6 shows that this sink shrinks as the grid is refined, for three reasons.

1. *Thinner clouds produce less rain.* Total rain production (autoconversion plus accretion) at a target N_d of 200 cm^{-3} falls from 0.55 mm d^{-1} at 40 km to 0.03 mm d^{-1} at 312 m (Fig. 6a). The cloud thins from about 650 m to about 300 m (Fig. 6d), and the KK2000 rates depend steeply on cloud water.

2. *More of the rain evaporates.* Cloud base rises, so the fraction of rain that evaporates below cloud increases, from 0.4 at 40 km to 0.93 at 312 m for the same N_d (Fig. 6b).

3. *Entrainment drying grows.* The entrainment drying flux $-\rho L_v w_\mathrm{e} \Delta q_t$ roughly doubles (Fig. 7c). As a result, the ratio of surface rain to entrainment drying, a measure of the relative importance of the two moisture sinks, falls by one to two orders of magnitude at fixed N_d (Fig. 6c).

At $\Delta x \geq 10$ km and $N_\mathrm{d} \leq 50$ cm^{-3} the surface rain sink is 30–65 % as large as entrainment drying. At 312 m and $N_\mathrm{d} \geq 100$ cm^{-3} it is below 0.3 %. The raining fraction of the domain (surface rain above 0.5 mm d^{-1}) is at most 0.1 % for $N_\mathrm{d} \geq 200$ cm^{-3} at $\Delta x \leq 1.25$ km. With almost no rain left to suppress, the precipitation-suppression pathway has nothing to act on, and the falling branch set by entrainment is exposed. This is the mechanism behind the inverted V in mixed-layer and LES studies (Wood, 2007; Hoffmann et al., 2020, 2024). Here, however, the location of its turning point is set by the grid.

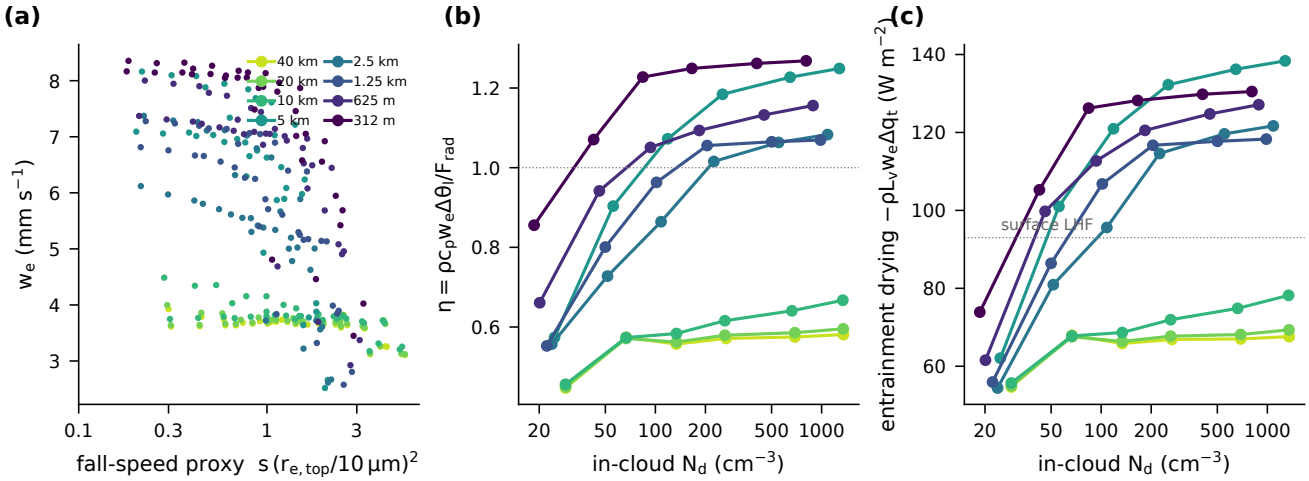


Figure 7. (a) Entrainment rate against a cloud-top fall-speed proxy for all runs with 3D turbulence off, coloured by grid spacing. (b) Ratio η of entrainment warming to boundary-layer radiative cooling. (c) Entrainment drying flux; the dotted line marks the prescribed surface latent heat flux. Panels (b) and (c): suppression on, sedimentation factor 1.

3.6 Entrainment and the boundary-layer budget

3.6.1 Entrainment rates and efficiency.

The entrainment rate at the default settings (target $N_d = 200\,\mathrm{cm}^{-3}$, suppression on) is 3.7 mm s^{-1} at 40 km, 6.8 mm s^{-1} at 2.5 km, 7.4 mm s^{-1} at 625 m and 8.1 mm s^{-1} at 312 m (Fig. 2c). Radiative cooling of the boundary layer is about 86 W m^{-2} in all runs with LWP above about 120 g m^{-2} (Table 3), so most of the change in w_e is a change in entrainment efficiency rather than in forcing. We define the ratio of entrainment warming to radiative cooling,

$$\eta = \frac{\rho c_p w_e \Delta\theta_l}{F_{rad}}. \tag{12}$$

At $\Delta x \geq 10$ km, η is 0.45–0.67 (Fig. 7b). At $\Delta x \leq 2.5$ km it exceeds one for $N_d \gtrsim 100$–$200\,\mathrm{cm}^{-3}$, and at 312 m it reaches 1.23–1.27 for $N_d \geq 100\,\mathrm{cm}^{-3}$. Entrainment warming then exceeds radiative cooling, and with the surface sensible heat flux the mixed layer warms by 0.07–0.10 K h^{-1} at 312 m. Likewise, entrainment drying (105–131 W m^{-2} at 312 m for $N_d \geq 50\,\mathrm{cm}^{-3}$) exceeds the surface latent heat flux of 93 W m^{-2}, whereas at $\Delta x \geq 10$ km it is 55–78 W m^{-2} (Fig. 7c). Across all runs, w_e decreases with a cloud-top fall-speed proxy $s\,(r_{e,top}/10\,\mu m)^2$ (Fig. 7a), as expected from the sedimentation–entrainment feedback (Bretherton et al., 2007). The decrease is much stronger and smoother at kilometre and sub-kilometre spacing. At $\Delta x \geq 10$ km, w_e varies by at most 1.3 mm s^{-1} over the full range of the proxy, mostly at 10 km.

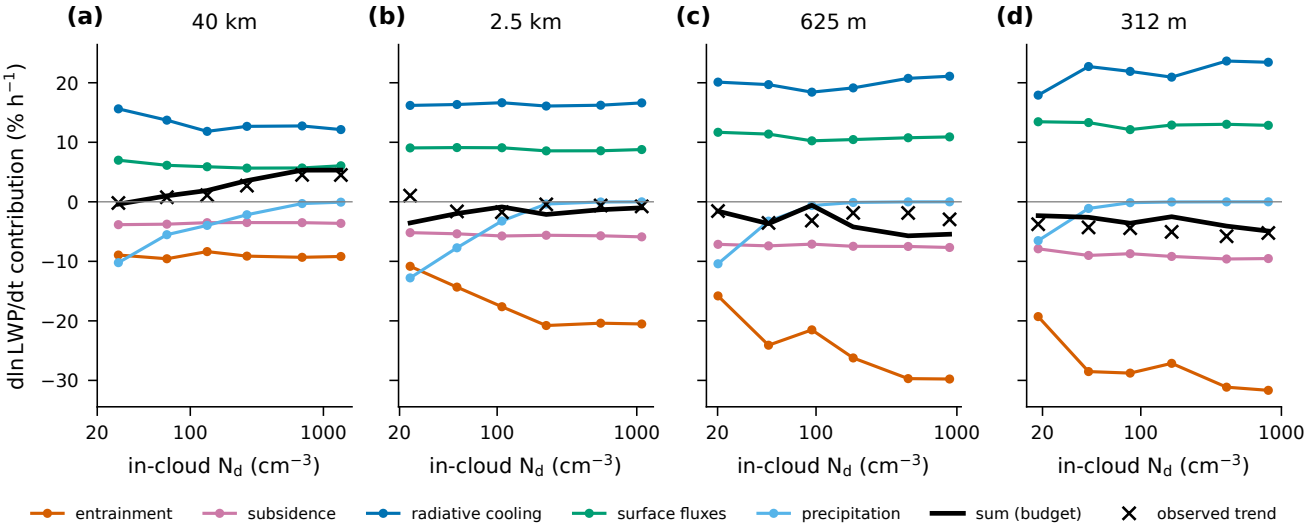


Figure 8. Mixed-layer LWP budget (Eq. 11) for hours 10–12, expressed as contributions to $d\ln\mathrm{LWP}/dt$, at (a) 40 km, (b) 2.5 km, (c) 625 m and (d) 312 m (suppression on, sedimentation factor 1). The black line is the budget sum; crosses are the simulated LWP trend.

3.6.2 LWP budget.

The mixed-layer LWP budget (Sect. 2.6) turns these fluxes into LWP tendencies (Fig. 8). In every run, radiative cooling and surface fluxes thicken the cloud, while entrainment and subsidence thin it. The balance shifts with grid spacing as follows (suppression on, $s = 1$, hours 10–12):

- *At 40 km*, the entrainment contribution is -8 to $-10\,\%\,\mathrm{h}^{-1}$ at all N_d. Precipitation contributes $-10\,\%\,\mathrm{h}^{-1}$ at $N_\mathrm{d} = 20\,\mathrm{cm}^{-3}$ and nearly zero at $N_\mathrm{d} \geq 500\,\mathrm{cm}^{-3}$. Removing the precipitation sink is what lets high-N_d clouds keep thickening: the budget sum rises from $-0.4\,\%\,\mathrm{h}^{-1}$ at $N_\mathrm{d} = 20\,\mathrm{cm}^{-3}$ to $+5\,\%\,\mathrm{h}^{-1}$ at $N_\mathrm{d} \geq 500\,\mathrm{cm}^{-3}$, matching the simulated trend.

- *At 312 m*, the entrainment contribution is -27 to $-32\,\%\,\mathrm{h}^{-1}$ for $N_\mathrm{d} \geq 50\,\mathrm{cm}^{-3}$, about three times as large. It becomes more negative with increasing N_d and is only partly offset by stronger radiative ($+18$ to $+24\,\%\,\mathrm{h}^{-1}$) and surface ($+12$ to $+13\,\%\,\mathrm{h}^{-1}$) contributions. Precipitation contributes less than $1\,\%\,\mathrm{h}^{-1}$ for $N_\mathrm{d} \geq 100\,\mathrm{cm}^{-3}$. The budget sum is negative at all N_d (-2.3 to $-4.9\,\%\,\mathrm{h}^{-1}$), consistent with the simulated thinning of -3.8 to $-5.8\,\%\,\mathrm{h}^{-1}$.

The resolution dependence of the LWP response is therefore, at root, a resolution dependence of entrainment efficiency. Its N_d dependence at fine spacing comes from the entrainment term, through sedimentation, rather than from precipitation.

3.7 Turbulence grey zone and boundary-layer structure

3.7.1 Resolved and parameterised variance.

At $\Delta x \geq 10\,\mathrm{km}$ all vertical-velocity variance is parameterised (Fig. 9a), and SHOC's boundary-layer variance is large (0.28–0.38 $\mathrm{m^2\,s^{-2}}$). Below 5 km, SHOC's variance collapses to about 0.07–0.11 $\mathrm{m^2\,s^{-2}}$ and keeps decreasing with Δx, to 0.03 $\mathrm{m^2\,s^{-2}}$ at 312 m, while the resolved variance grows. The resolved share is 3–7 % at 5 km, 4–9 % at 2.5 km, 9–19 % at 1.25 km, 28–38 % at 625 m and 54–67 % at 312 m. Even at 312 m, a third of the variance is still subgrid. Total variance at kilometre spacing is three to four times smaller than at the coarse spacings, yet entrainment is about twice as large. The partly resolved circulation entrains more efficiently per unit variance than SHOC does in the single-column limit.

3.7.2 Sc-like and Cu-like regimes.

The resolved vertical-velocity skewness (Fig. 9b) separates two regimes.

- At low N_d it is positive (up to 1.9 at 312 m for $N_\mathrm{d} = 20\,\mathrm{cm^{-3}}$): narrow, strong updrafts in a drizzle-decoupled, cumulus-like boundary layer.
- At high N_d and 1.25–2.5 km it is negative (-0.7 to -1.0), the signature of radiatively driven downdrafts in a well-mixed stratocumulus layer (Wood, 2012). The transition occurs at $N_\mathrm{d} \approx 50$–$100\,\mathrm{cm^{-3}}$. At 312 m the skewness stays positive (0.09–0.24 for $N_\mathrm{d} \geq 100\,\mathrm{cm^{-3}}$), so the finest boundary layer is structurally different again.

3.7.3 Decoupling.

Drizzle decouples the boundary layer at low N_d (Fig. 9c). At $\Delta x \leq 5$ km, Δq_t is 1.1–1.6 $\mathrm{g\,kg^{-1}}$ at $N_\mathrm{d} = 20\,\mathrm{cm^{-3}}$ and decreases to 0.2–0.6 $\mathrm{g\,kg^{-1}}$ at $N_\mathrm{d} = 1000\,\mathrm{cm^{-3}}$, consistent with below-cloud evaporation stabilising the subcloud layer (Stevens et al., 1998; Bretherton and Wyant, 1997; Jones et al., 2011). The single-column limit has a structurally different profile: q_t decreases upward by 1.2–1.7 $\mathrm{g\,kg^{-1}}$ at all N_d, while θ_l is unstably stratified for $N_\mathrm{d} \geq 100\,\mathrm{cm^{-3}}$ across a deep SHOC-mixed cloud layer.

3.7.4 Cell scale.

The dominant LWP wavelength converges faster than LWP itself (Fig. 9d; Appendix Fig. D1). At 5 km the cells fill the domain (48 km is one fifth of the domain), and the aspect ratio λ/z_i falls from 33 at 2.5 km to 13–20 at 625 m and 312 m. This is within the range of aspect ratios of observed closed mesoscale cells (Wood and Hartmann, 2006) and of LES with large domains (Kazil et al., 2017; Zhou et al., 2018; Zhou and Bretherton, 2019).

3.7.5 3D turbulence.

Switching on 3D turbulence changes LWP by less than $1.5\,\mathrm{g\,m^{-2}}$ ($< 1.5\,\%$), cloud cover by less than 0.007 and w_e by -0.13 to $+0.02\,\mathrm{mm\,s^{-1}}$ at both 625 m and 312 m (Appendix C). It is not a factor in the results above.

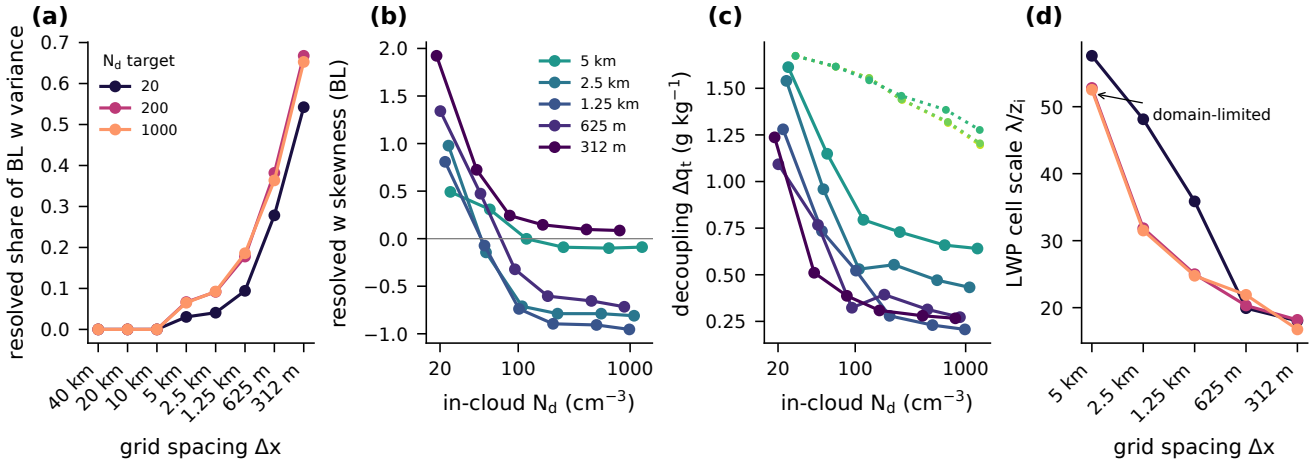


Figure 9. Boundary-layer structure (suppression on, sedimentation factor 1). (a) Resolved share of boundary-layer vertical-velocity variance against grid spacing for three target N_d. (b) Resolved vertical-velocity skewness over the boundary layer and (c) decoupling Δq_t against in-cloud N_d (dotted: $\Delta x \geq 10\,\mathrm{km}$). (d) Dominant LWP wavelength relative to z_i; at 5 km the scale is limited by the domain.

3.8 Convergence

None of the bulk quantities converges with grid spacing (Fig. 10). From 625 to 312 m, LWP decreases by a further 24 % at $N_\mathrm{d} = 20\,\mathrm{cm}^{-3}$ and 36–40 % at $N_\mathrm{d} \geq 100\,\mathrm{cm}^{-3}$, compared with 12–23 % from 1.25 km to 625 m. Cloud cover decreases by 10–18 % and cloud thickness by 7–20 %. For LWP, the ratio of the change over the last halving to the change over the previous one exceeds one (1.3–2.0) for all $N_\mathrm{d} \geq 50\,\mathrm{cm}^{-3}$. The sequence is not in an asymptotic range, and a Richardson-type extrapolation is not meaningful. The entrainment rate is non-monotonic: it dips at 2.5–1.25 km and rises again below. This is a typical grey-zone signature, as the parameterised and resolved contributions trade places. In contrast to the bulk quantities, the clean-branch susceptibility (0.65–0.76 at $\Delta x \leq 2.5\,\mathrm{km}$) and the location of the cell-scale peak change comparatively little at sub-kilometre spacing. We stress, however, that the sign of $S_{100\to200}$ at 312 m could still change with further refinement, and that the vertical grid (about 50 m at the inversion) is held fixed. Vertical resolution at the inversion is known to control entrainment in LES and in SCREAM (Stevens et al., 2005; Bogenschutz et al., 2023b).

3.9 Cloud-top retrievals and within-scene covariability

Satellite estimates of LWP adjustments use cloud-top N_d retrieved from cloud-top effective radius and optical depth (Grosvenor et al., 2018). The model's AeroCom-style cloud-top N_d is 44–72 % of the in-cloud layer value at $\Delta x \geq 10\,\mathrm{km}$ and 77–95 % at 312 m. Because the ratio varies little with N_d at a given spacing, susceptibilities computed with cloud-top N_d agree closely with the layer-based values (Fig. 11a). The maximum difference is 0.10 on the clean branch and 0.04 and 0.01 on the moderate and polluted branches.

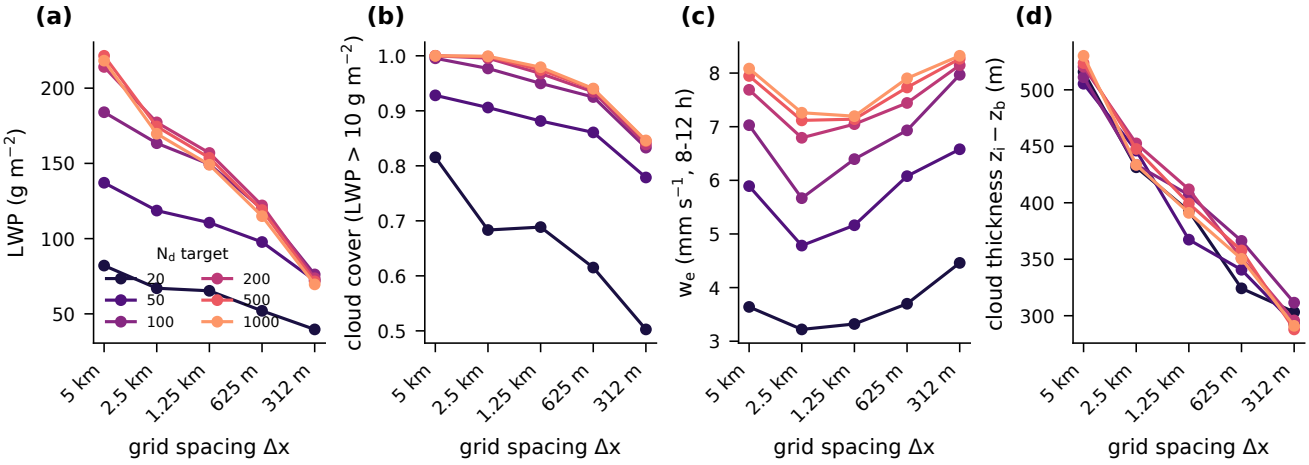


Figure 10. Hour 10–12 means against grid spacing ($\Delta x \leq 5\,\mathrm{km}$) for each target N_d (suppression on, sedimentation factor 1). (a) LWP, (b) cloud cover, (c) entrainment rate and (d) cloud thickness.

Within each domain, CCN are horizontally uniform, so any spatial covariation of LWP with cloud-top N_d arises from cloud processes rather than aerosol. We regress column $\ln\mathrm{LWP}$ on column cloud-top $\ln N_\mathrm{d}$ over cloudy columns at each 15 min instant in hours 10–12. The resulting within-scene slopes span -16 to $+3$ (Fig. 11c), with correlations of -0.95 to $+0.6$. Their sign depends on spacing and N_d:

- At $\Delta x \geq 2.5\,\mathrm{km}$ and low N_d they are strongly negative. Collision–coalescence depletes N_d in the thickest, drizzling columns (Fig. 11b).
- At $\Delta x \leq 625\,\mathrm{m}$ and $N_\mathrm{d} \geq 50\,\mathrm{cm}^{-3}$ they are positive (0.4–3.3). Thin, evaporating cloud edges have reduced cloud-top N_d.

In no case does the within-scene slope resemble the across-run susceptibility. At 312 m and target $N_\mathrm{d} = 200\,\mathrm{cm}^{-3}$, for example, the within-scene slope is $+1.9$ while $S_{100\to200} = -0.05$. This supports the caution that spatial LWP–N_d relationships reflect cloud-controlled N_d variability as well as aerosol-controlled variability (Gryspeerdt et al., 2019; Arola et al., 2022; Mülmenstädt et al., 2024b). It also suggests that the sign of such spurious covariability can itself depend on model resolution.

4 Discussion

4.1 A grid-controlled regime shift

The results describe a regime shift in the dominant adjustment pathway, controlled by grid spacing. In the single-column limit ($\Delta x \geq 10\,\mathrm{km}$), SHOC's entrainment hardly responds to droplet sedimentation, and the clouds are thick enough to rain substantially. The LWP response is then positive, monotonic and due entirely to precipitation suppression. That is the classic

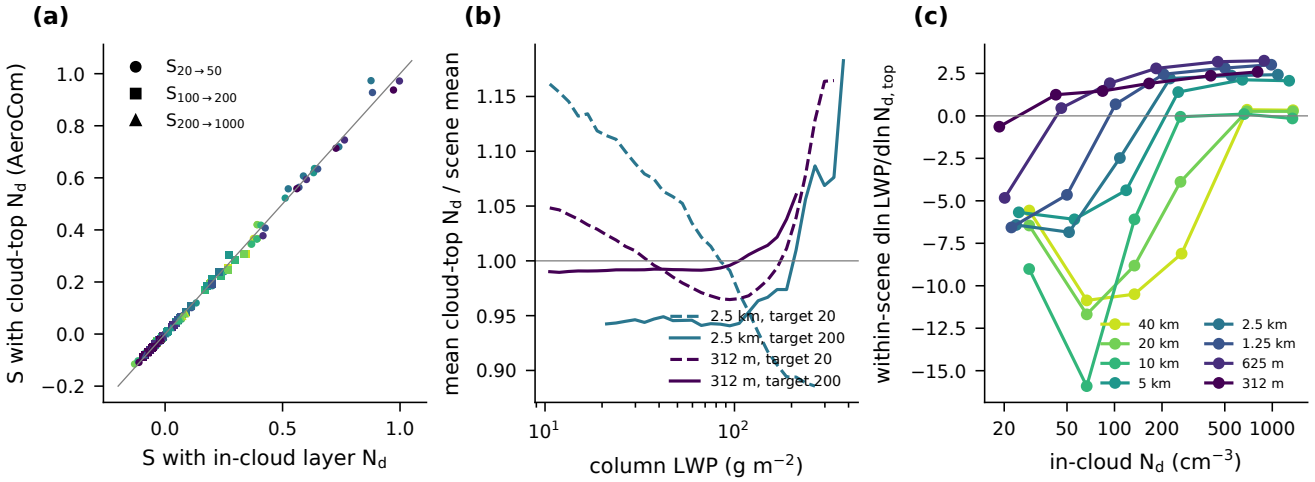


Figure 11. (a) Susceptibility computed with AeroCom-style cloud-top N_d against that computed with in-cloud layer N_d, for all branches and configurations (3D turbulence off). (b) Mean cloud-top N_d (relative to the scene mean) as a function of column LWP, for 2.5 km and 312 m at target N_d of 20 and 200 cm^{-3} (hours 10–12). (c) Within-scene regression slope of column $\ln \mathrm{LWP}$ on column cloud-top $\ln N_\mathrm{d}$ (suppression on, sedimentation factor 1).

Albrecht (1989) picture, and broadly the behaviour of GCMs (Quaas et al., 2009; Michibata et al., 2016; Mülmenstädt et al., 2020).

As the boundary-layer eddies begin to be resolved, two things happen together. First, entrainment becomes more efficient: η rises above one and the boundary layer warms, dries and deepens. The cloud thins and its base rises until little rain is produced and most of what forms evaporates. Second, entrainment becomes sensitive to droplet sedimentation, so polluted clouds entrain more than clean ones. The first effect removes the leverage of precipitation suppression; the second supplies a negative entrainment pathway. The two effects together move the turning point of the inverted V from beyond 1000 cm^{-3} to about 100 cm^{-3}.

This picture is consistent with mixed-layer and LES arguments for a turning point determined by the competition between precipitation and entrainment time scales (Wood, 2007; Hoffmann et al., 2020, 2024). Our contribution is to show that, in a model that spans the grey zone, the competition is tilted by the grid itself.

4.2 Implications for storm-resolving and climate models

Most GSRMs run at 2.5–5 km (Stevens et al., 2019; Caldwell et al., 2021). In EAMxx at this spacing, $S_{100\to200}$ is still positive (0.11–0.20 with suppression on), but only about a third to two thirds of the 40 km value. The polluted branch is near zero or slightly negative. The 2.5 km susceptibility is thus neither the GCM-like answer nor the sub-kilometre answer. It reflects a model whose entrainment is partly resolved and whose subgrid scheme has not been designed for this partition. That such a GSRM simulates a weaker or even negative LWP adjustment than a GCM (cf. Sato et al., 2018; Terai et al., 2020) should not,

by itself, be taken as evidence that the GSRM is physically more realistic. In our simulations, LWP decreases by 20–40 % with each halving of the grid spacing below 1.25 km, with no sign of convergence.

For GCMs, the coarse runs show that the precipitation-suppression pathway can produce positive susceptibilities of about 0.3 by itself. They also show that SHOC-type entrainment in the single-column limit hardly responds to N_{d}. GCMs that lack an explicit sedimentation–entrainment coupling would be expected to miss the negative pathway in the same way. This agrees with the finding that GCMs simulate negative present-day LWP–N_{d} correlations but still positive LWP responses to anthropogenic aerosol (Mülmenstädt et al., 2024b), and with the difficulty of representing entrainment-driven adjustments in GCMs (Mülmenstädt et al., 2024a; Karset et al., 2020).

The factorial design suggests a constraint for parameterisation development: a scheme should reproduce both the contribution of precipitation suppression (S_{ps}) and the sensitivity to sedimentation (dS/ds) found in a converged reference. That reference does not yet exist for this model.

4.3 Relation to satellite-based estimates

The cloud-top retrieval proxy reproduces the layer-based susceptibilities closely in these idealised scenes (Fig. 11a). Sampling cloud tops does not by itself bias S here. However, within-scene LWP–N_{d} covariation with uniform aerosol spans an order of magnitude more than the true susceptibility and changes sign with grid spacing (Fig. 11c). This adds a model-based illustration to the concerns raised by Gryspeerdt et al. (2019) and Arola et al. (2022): spatial co-variability is not a measure of the adjustment. It also cautions against evaluating the LWP adjustments of kilometre-scale models with satellite statistics that do not control for cloud-driven N_{d} variability.

4.4 Limitations

The following limitations bear on how far the results generalise.

1. *Single case and forcing.* We simulate one nocturnal, drizzling case (RF02) with constant forcing and no shortwave radiation. The diurnal cycle modulates adjustments (Sandu et al., 2008), and the humidity above the inversion controls the sign of the entrainment response (Ackerman et al., 2004; Chun et al., 2023). The free troposphere of RF02 is relatively moist, so the negative pathway may be stronger in drier cases.

2. *Single realisation.* Each configuration was run once. The quoted uncertainty reflects temporal sampling, not initial-condition spread. Small negative susceptibilities at sub-kilometre spacing (-0.02 to -0.08) should be confirmed with ensembles.

3. *Transience.* After 12 h the sub-kilometre runs are still thinning by 3–6 % h^{-1}, and their high-N_{d} susceptibilities are still becoming more negative (Fig. 4). Longer simulations would be needed to assess the equilibrium response (Glassmeier et al., 2021; Bretherton et al., 2010).

4. *Resolution.* The finest grid (312 m, physics columns at 469 m) is far coarser than LES of RF02 (Ackerman et al., 2009), and the vertical spacing at the inversion (about 50 m) is fixed. Neither LWP nor $S_{100\to200}$ is converged. The results at 312 m should be read as the trend of a grey-zone model, not as a converged answer.

5. *Domain.* The 240 km domain holds 12–18 cells at sub-kilometre spacing but limits the cell scale at 5 km. The coarse runs are effectively single columns.

6. *Aerosol.* CCN are prescribed and uniform. There is no aerosol processing, scavenging or replenishment, which can matter for drizzle-driven transitions (Wood and Hartmann, 2006; Berner et al., 2013; Goren et al., 2019; Yamaguchi et al., 2017). Diagnosed N_{d} differs from the target by up to 64 %; we account for this by using diagnosed N_{d}.

7. *Model physics.* Condensation and evaporation are done by SHOC's saturation adjustment, so the droplet-size dependence of cloud-top evaporation, which LES identify as a contributor to the entrainment pathway (Wang and Feingold, 2009; Xue and Feingold, 2006; Small et al., 2009), is absent. The autoconversion scheme is KK2000, which was fitted to LES of stratocumulus. Other schemes would change S_{ps} (Morrison et al., 2020).

8. *Diagnostics.* Surface precipitation includes the cloud-droplet sedimentation flux. P3's sedimentation-tendency diagnostics are unusable in the version used here (Appendix A), so we could not diagnose the cloud-top sedimentation flux directly and used a fall-speed proxy instead.

5 Conclusions

We ran 358 doubly periodic EAMxx simulations of the DYCOMS-II RF02 drizzling stratocumulus, spanning grid spacings from 40 km to 312 m and N_{d} from 20 to 1000 cm^{-3}, with factorial switches for precipitation suppression and droplet sedimentation. The main conclusions are as follows.

1. *The LWP–N_{d} relationship depends strongly on grid spacing.* At 10–40 km, LWP increases monotonically with N_{d} ($S_{100\to200} = 0.24$–0.27). With refinement the maximum moves to lower N_{d}, reaching about 100 cm^{-3} at 312 m, where $S_{100\to200} = -0.05$ and $S_{200\to1000} = -0.04$. An inverted V emerges only once boundary-layer eddies are partly resolved.

2. *The controlling pathway changes.* Precipitation suppression explains all of the positive response at coarse spacing. Its contribution on the moderate branch falls from 0.30 at 40 km to 0.02 at 312 m, and the precipitation-suppression gain in LWP at $N_{\mathrm{d}} = 1000\,\mathrm{cm}^{-3}$ falls from 134 to 6 $\mathrm{g\,m}^{-2}$. At sub-kilometre spacing the falling branch persists without precipitation suppression and steepens with faster droplet sedimentation. It is entrainment-driven.

3. *Entrainment efficiency is the root cause.* The ratio of entrainment warming to radiative cooling rises from about 0.5 at coarse spacing to 1.2–1.3 at 312 m. Entrainment drying exceeds the surface moisture supply, and the clouds thin, lift and rain less. A mixed-layer LWP budget reproduces the simulated tendencies (correlation 0.88). It attributes the thinning, and its N_{d} dependence at fine spacing, to the entrainment term.

4. *The clean branch is robust; bulk quantities are not converged.* $S_{20\to50}$ is 0.65–0.76 at all spacings of 2.5 km and finer, much of it from cloud-cover changes. LWP, cloud cover and cloud thickness still change by 10–40 % between 625 and 312 m, and 3D turbulence makes no appreciable difference.

5. *Satellite-style diagnostics behave differently from the susceptibility.* Cloud-top N_{d} gives the same susceptibilities as layer N_{d}. Within-scene LWP–N_{d} slopes with uniform aerosol span -16 to $+3$ and change sign with grid spacing.

For GSRMs at 1–5 km, these results imply that the simulated aerosol effect on stratocumulus LWP is controlled by the partition between resolved and parameterised turbulence, as much as by the microphysics. Interpreting GSRM aerosol forcing will require grey-zone-aware turbulence closures or explicit convergence tests in LES-like configurations. Ensembles, longer and diurnally forced simulations, additional cases and finer vertical and horizontal grids are the natural next steps.

Appendix A: A diagnostic bug in the P3 sedimentation tendencies

P3's sedimentation-tendency outputs `qc_sed`, `qr_sed` and `qi_sed` cannot be used in the EAMxx version used here. The C++ sedimentation routines compute the tendency as $(q - q_{\mathrm{tend}})/\Delta t$, assuming that the tendency buffer holds the pre-sedimentation mixing ratio on entry. In the Fortran implementation the buffer is initialised that way (`qc_sed = qc` before the call). The C++ driver never loads it, so the output is approximately $q/\Delta t$; for example, `qc_sed`/`qc` equals $1/(60\,\mathrm{s})$ at 1.25 km. The model state is not affected. The fix is to copy q_{c}, q_{r} and q_{i} into the three buffers before the sedimentation calls. Because of this bug we characterise sedimentation by the fall-speed proxy of Fig. 7a.

Appendix B: Mixed-layer budget validation

Figure B1 compares the mixed-layer budget predictions of Sect. 2.6 with the simulated tendencies over all 2 h windows from 2 to 12 h, for the 288 runs with 3D turbulence off. The θ_l and q_t tendencies are reproduced with correlations of 0.89 and 0.87. The largest departures are at 5 km, where the boundary layer is strongly decoupled and the cloud base departs most from the mixed-layer lifting condensation level. LWP tendencies are reproduced with a correlation of 0.88 and a mean bias of $-1.1\,\mathrm{g\,m^{-2}\,h^{-1}}$. Using the observed dz_{b}/dt instead of the LCL-based estimate degrades the agreement, because z_{b} is quantised by the vertical grid.

Appendix C: 3D turbulence

Figure C1 shows the difference between runs with 3D turbulence on and off, for the 36 pairs at 625 m and 34 of the 36 pairs at 312 m. Changes in LWP are between -1.0 and $+1.4\,\mathrm{g\,m^{-2}}$, changes in cloud cover between -0.005 and $+0.007$, and changes in the entrainment rate between -0.13 and $+0.02\,\mathrm{mm\,s^{-1}}$. The only systematic signal is a slight reduction in entrainment. These differences are an order of magnitude smaller than the differences between successive grid spacings or between N_{d} levels.

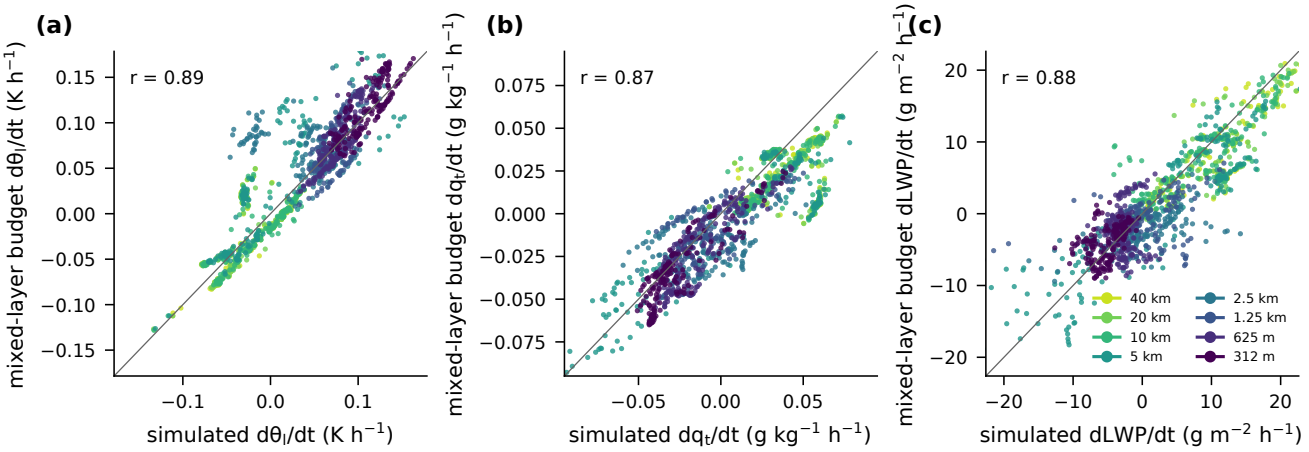


Figure B1. Mixed-layer budget against simulated tendencies of (a) mixed-layer θ_l, (b) mixed-layer q_t and (c) LWP, for all 2 h windows and all runs with 3D turbulence off, coloured by grid spacing.

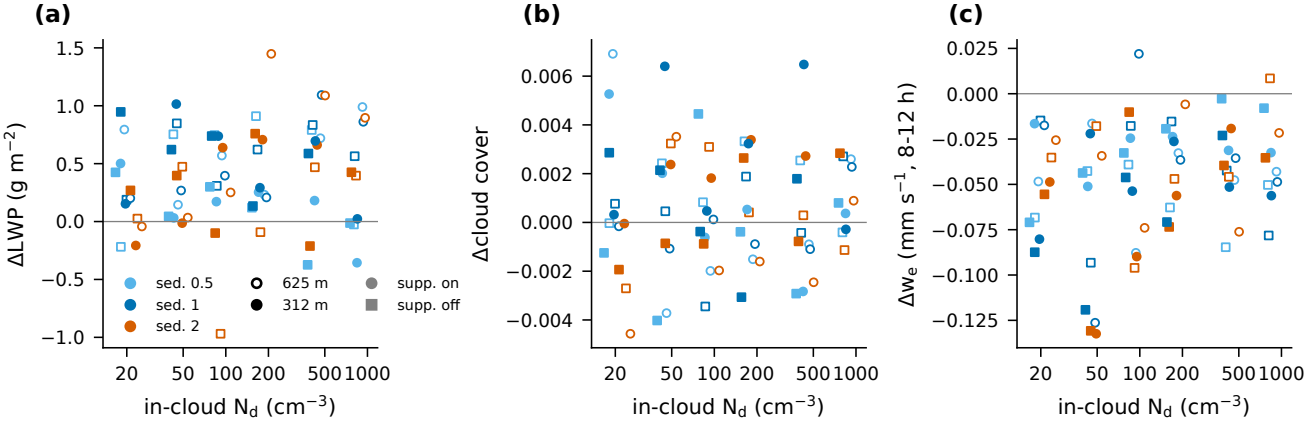


Figure C1. Difference (3D turbulence on minus off) in (a) LWP, (b) cloud cover and (c) entrainment rate at 625 m (open symbols) and 312 m (filled), coloured by sedimentation factor; circles: suppression on, squares: off.

Appendix D: LWP spectra and distributions

Figure D1 shows the LWP spectra and column-LWP distributions behind Sect. 3.7. The spectral peak shifts from 48 km at 5 km (the domain limit) to about 30 km at 2.5 km and to 15–20 km at 625 and 312 m, and the spectra at the two finest grids peak at similar wavelengths. The LWP distributions broaden and shift to lower values as the grid is refined. At $N_\mathrm{d} = 20\,\mathrm{cm}^{-3}$ they develop a large population of clear or nearly clear columns.

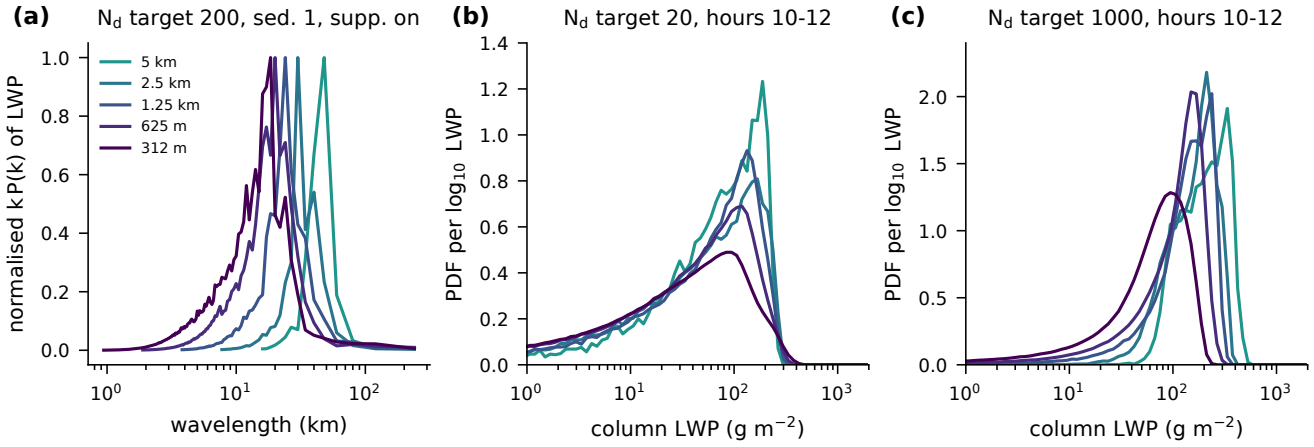


Figure D1. (a) Normalised radially averaged LWP spectra ($k\,P(k)$, hours 10–12) for target $N_\mathrm{d} = 200\,\mathrm{cm}^{-3}$. (b, c) Distributions of column LWP over hours 10–12 for target N_d of 20 and $1000\,\mathrm{cm}^{-3}$. Suppression on, sedimentation factor 1. In (b), clear columns at small LWP are off scale.

Code and data availability. The model code is the E3SM branch `mahf708/eamxx/p3-cloud-sed-factor` (https://github.com/mahf708/E3SM/tree/mahf708/eamxx/p3-cloud-sed-factor). It contains E3SM pull request #9017 (the AeroCom cloud-top output operator) and the P3 sedimentation-factor option. Run scripts, analysis scripts (including all scripts that produce the tables and figures of this paper) and the derived tables are in the `sed-precsupp-study` directory of the scmlib repository (https://github.com/mahf708/scmlib/tree/sed-precsupp-study). Model output (about 5 TB) currently resides on NERSC Perlmutter; the derived per-run tables and processed fields used here will be archived in a public repository before publication. [Authors: add DOI.]

Author contributions. NM and JM conceived the study. NM implemented the model modifications, designed and ran the simulations. The analysis and the first draft of the manuscript were produced with an AI assistant (Claude Code, Anthropic) working under NM's direction. Both authors must review and edit the manuscript. [Authors: confirm or revise.]

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